

Notes: Landais (AEJ:EP, 2015)

Assessing the Welfare Effects of Unemployment Benefits Using the Regression Kink Design

Contents

1	What the paper is doing	2
2	All models / key equations in the paper	2
2.1	Institutional model: UI schedules (benefit level and potential duration)	2
2.2	Two-period dynamic search model (baseline theoretical model)	3
2.3	Stochastic wage offers / reservation wage extension (McCall-type)	4
2.4	Multi-period generalization: Proposition 1 (key identification formula)	4
2.5	Empirical model: Regression Kink Design (RKD)	5
2.6	Mechanical vs behavioral component of duration extensions (used in interpretation)	5
3	Figures	6
4	Tables	9
5	Big picture summary	15
6	Conclusion	15
6.1	What is the paper ultimately concluding?	15
6.2	Summary of the paper (what it does, how, and what it finds)	16
6.3	Intuition	17

1 What the paper is doing

The paper proposes an empirical strategy to separate:

- **Moral hazard effects** of unemployment insurance (UI): the distortionary substitution response of search effort to UI generosity (Frisch-type effect; response holding marginal utility of wealth constant),
- **Liquidity effects**: the response of search effort to cash-on-hand / wealth (insurance / consumption-smoothing channel).

Key idea: exploit **time-profile variation** of UI benefits using two margins that move benefits at different times:

- **Benefit level** (weekly benefit amount) shifts benefits early in the spell.
- **Potential duration** pushes benefits later in the spell (extensions).

Comparing how *initial* search/exit responds to these two changes identifies moral hazard vs liquidity, after adjusting for state dependence.

Empirically, the paper uses a **Regression Kink Design (RKD)** based on kinked UI schedules in U.S. states (with administrative CWBH data).

2 All models / key equations in the paper

2.1 Institutional model: UI schedules (benefit level and potential duration)

Weekly benefit amount (benefit level). Let hqw be the claimant's *highest quarter of earnings* in the base period. UI weekly benefit amount is a deterministic kinked function:

$$b(hqw) = \begin{cases} \tau_1 \cdot hqw, & \text{if } \tau_1 \cdot hqw \leq b_{\max}, \\ b_{\max}, & \text{if } \tau_1 \cdot hqw > b_{\max}. \end{cases} \quad (1)$$

So the kink occurs at k_b such that $\tau_1 k_b = b_{\max}$.

Total benefits ceiling and potential duration. Let bpw be total base-period earnings. Let $D_{\max}$ be the maximum weeks allowed (often 26), and τ_2 the fraction mapping base-period earnings into a *total benefit cap*.

$$B = \min\{D_{\max} \cdot b, \tau_2 \cdot bpw\}. \quad (2)$$

Potential duration (weeks of eligibility) is

$$D = \frac{B}{b} = \min \left\{ D_{\max}, \tau_2 \cdot \frac{bpw}{\min(\tau_1 hqw, b_{\max})} \right\}. \quad (3)$$

In many cases, the relevant assignment variable for duration is the ratio bpw/hqw (kinked mapping into D).

2.2 Two-period dynamic search model (baseline theoretical model)

A worker is unemployed at the start of period 0. If unemployed at the start of period $t \in \{0, 1\}$, she chooses search effort s_t with cost $\psi(s_t)$, $\psi' > 0$, $\psi'' > 0$. Search effort maps into a probability of employment (hazard).

Preferences are time-separable with CRRA-like utility $u(\cdot)$, $u' > 0$, $u'' < 0$. (The paper keeps this generic.)

Let initial assets be A_0 , assets after period 0 be A_1 , and terminal assets $\bar{A} \geq 0$. Let wage be w_t if employed in period t , payroll tax τ , and benefits b_t if unemployed in t .

Consumption in each state:

$$c_0^e = A_0 - A_1 + w_0 - \tau, \quad c_0^u = A_0 - A_1 + b_0, \quad (4)$$

$$c_1^e = A_1 - \bar{A} + w_1 - \tau, \quad c_1^u = A_1 - \bar{A} + b_1. \quad (5)$$

Lifetime utility at start of period 0:

$$U = s_0 u(c_0^e) + (1 - s_0) u(c_0^u) - \psi(s_0) + \beta \left(s_0 u(c_1^e) + (1 - s_0) (s_1 u(c_1^e) + (1 - s_1) u(c_1^u) - \psi(s_1)) \right). \quad (6)$$

(1) First-order condition for search effort at $t = 0$

$$\psi'(s_0) = \underbrace{u(c_0^e) + \beta u(c_1^e)}_{\text{lifetime utility if employed in period 0}} - \underbrace{\left(u(c_0^u) + \beta (s_1 u(c_1^e) + (1 - s_1) u(c_1^u) - \psi(s_1)) \right)}_{\text{lifetime utility if unemployed in period 0}}. \quad (1)$$

(2) Decomposition of effect of current benefits b_0

$$\frac{\partial s_0}{\partial b_0} = - \frac{u'(c_0^u)}{\psi''(s_0)} = \frac{\partial s_0}{\partial A_0} - \frac{\partial s_0}{\partial w_0}. \quad (2)$$

Interpretation of (2). Increasing current UI benefits b_0 reduces search effort. The change can be viewed as:

- a **wealth/liquidity effect** ($\partial s_0 / \partial A_0$): more assets/cash-on-hand reduces the urgency to search,
- a **moral hazard (Frisch/substitution) effect** ($-\partial s_0 / \partial w_0$): benefits lower the net return to working/searching today.

(3)–(4) Euler equations (optimal intertemporal allocation)

$$u'(c_0^e) = \beta u'(c_1^e), \quad (3)$$

$$u'(c_0^u) = \beta (s_1 u'(c_1^e) + (1 - s_1) u'(c_1^u)). \quad (4)$$

(5) State dependence: effect of future wage on current effort

$$\frac{\partial s_0}{\partial w_1} = (1 - s_1) \cdot \frac{\partial s_0}{\partial w_0}. \quad (5)$$

Interpretation of (5). Future wages matter less for current effort because:

- If I search now, I may benefit from w_1 tomorrow if employed.
- But I can also postpone search to period 1; the ability to do that reduces the marginal value of increasing w_1 for effort today.

The scaling factor $(1 - s_1)$ comes from state dependence (today's effort affects whether I'm unemployed in period 1).

(6) Decomposition of effect of future benefits b_1

$$\frac{\partial s_0}{\partial b_1} = -\beta \frac{(1 - s_1)u'(c_1^u)}{\psi''(s_0)} = \frac{\partial s_0}{\partial A_0} - (1 - s_1) \frac{\partial s_0}{\partial w_0}. \quad (6)$$

Interpretation of (6). Unlike a model without state dependence, future benefits b_1 affect current effort not only via wealth (marginal utility) but also via the return-to-search channel.

(7) Comparing current vs future benefit effects

$$\frac{\partial s_0}{\partial b_0} - \frac{\partial s_0}{\partial b_1} = -s_1 \cdot \frac{\partial s_0}{\partial w_0}. \quad (7)$$

Interpretation of (7). The difference between how effort responds to benefits today vs benefits tomorrow identifies the moral hazard effect up to a factor involving s_1 (again: state dependence).

2.3 Stochastic wage offers / reservation wage extension (McCall-type)

If job offers arrive stochastically and acceptance depends on a reservation wage R_t , the two-period comparison becomes:

$$\frac{\partial s_0}{\partial b_0} - \frac{\partial s_0}{\partial b_1} = -h_1 \cdot \frac{\partial s_0}{\partial w_0}, \quad (9)$$

where

$$h_1 = s_1 \Pr[w \geq R_1]$$

is the *hazard rate out of unemployment* in period 1.

Interpretation of (9). Even with reservation wages, the same logic goes through: the “future-benefit” response differs from the “current-benefit” response by a factor proportional to the period-1 exit hazard.

2.4 Multi-period generalization: Proposition 1 (key identification formula)

Let UI provide weekly benefits b for B periods (potential duration). Assume borrowing constraint does not bind after B periods.

Define:

- $S_1(B)$: survival rate at time B conditional on being unemployed at period 1,

- $\bar{S}_1^B$: average survival rate between time 1 and time B conditional on being unemployed at period 1,
- D_1^B : average duration of *covered* UI spells conditional on being unemployed at time 1,
- Θ_1 : moral hazard effect (Frisch-type) of providing UI benefits b for B periods.

The paper's key identification relationship:

$$\frac{1}{B} \frac{\partial s_0}{\partial b} \Big|_B - \frac{1}{b} \frac{\partial s_0}{\partial B} = - \frac{\bar{S}_1^B - S_1(B)}{D_1^B} \cdot \Theta_1. \quad (8)$$

Interpretation of Proposition 1 / (8). A benefit-level increase vs a benefit-duration increase can be scaled to deliver the *same total dollar liquidity* to the unemployed (heuristic: $B db = b dB$). The difference between the *initial exit* responses, properly scaled, isolates moral hazard after correcting for state dependence via survival objects.

2.5 Empirical model: Regression Kink Design (RKD)

Let W be the assignment (running) variable and k the kink location. Define $D = \mathbf{1}[W \geq k]$. Local polynomial RKD regression:

$$\mathbb{E}[Y \mid W = w] = \mu_0 + \sum_{p=1}^{\bar{p}} (\gamma_p (w - k)^p + \nu_p (w - k)^p \cdot D), \quad |w - k| \leq h. \quad (10)$$

RKD estimand (what I'm extracting). The kink estimand is the change in slope at the kink:

$$\text{RKD slope change} = \nu_1.$$

Since the treatment rule's slope change is deterministic, the RKD treatment effect is the ratio:

$$\widehat{\text{ATE}}_{\text{local}} = \frac{\hat{\nu}_1}{\Delta(\text{slope of treatment at kink})}.$$

In the paper's tables this shows up as α (for benefit level) and β (for potential duration).

2.6 Mechanical vs behavioral component of duration extensions (used in interpretation)

Let $S(t)$ be the survival function in unemployment up to time t . If benefits exhaust at B ,

$$D_B = \sum_{t=0}^{B-1} S(t).$$

Then a one-week increase in B changes average UI-claims duration via:

$$\frac{dD_B}{dB} = \sum_{t=0}^{B-1} \frac{dS(t)}{dB} + S(B).$$

Takeaway: the observed effect of extending potential duration contains:

- a **mechanical** part $S(B)$ (some people are still unemployed at exhaustion, so pushing exhaustion out by 1 week adds duration mechanically),
- a **behavioral** part $\sum_{t=0}^{B-1} dS(t)/dB$ (people search less, delaying exits).

3 Figures

Figure 1: Backing out moral hazard in dynamic labor supply / dynamic UI

- Panel A (labor supply analogy): a permanent tax change shifts the whole net wage profile $\Rightarrow$ labor response mixes wealth and Frisch (substitution) effects. Comparing a change in wages today vs tomorrow can isolate Frisch effects because future wages influence today mainly through wealth (in the no-state-dependence benchmark).
- Panel B (UI setting): increasing benefit level db shifts benefits early; increasing potential duration dB shifts benefits later. If scaled so $B db = b dB$, the liquidity injected is comparable.
- Key twist: **state dependence** (today's search affects tomorrow's state) means future-benefit changes also affect today's incentives, so the comparison identifies moral hazard only after adjusting for survival terms (Proposition 1).

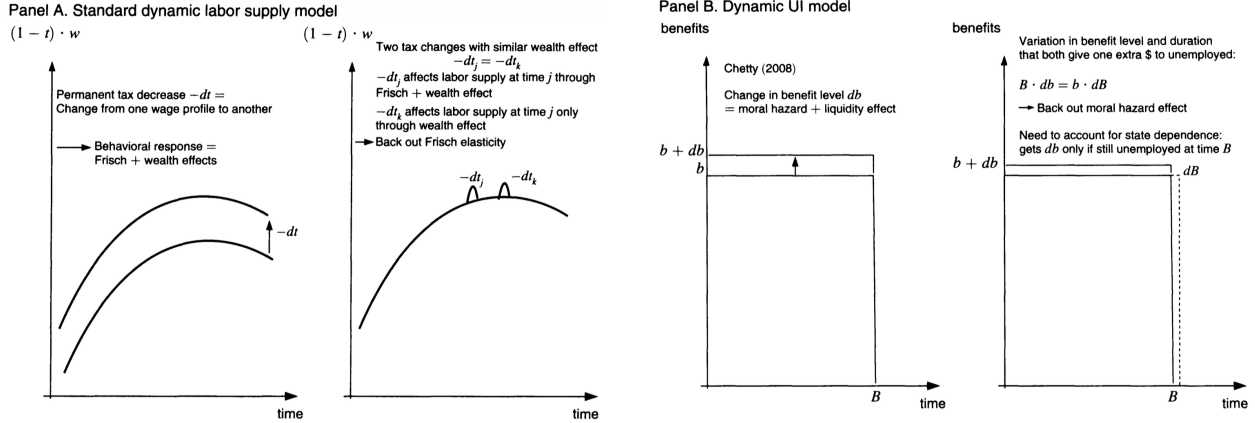


FIGURE 1. BACKING OUT MORAL HAZARD EFFECTS IN DYNAMIC LABOR SUPPLY MODELS

FIGURE 1. BACKING OUT MORAL HAZARD EFFECTS IN DYNAMIC LABOR SUPPLY MODELS (Continued)

Figure 2: Louisiana UI weekly benefit schedule (WBA) over time

- Shows the deterministic schedule mapping highest quarter earnings (hqw) into WBA.
- Linear up to the maximum benefit amount $b_{\max}$, then flat.
- The kink location moves over time because $b_{\max}$ changes (inflation adjustments, policy).
- Interpretation: these moving kinks create quasi-experimental variation suitable for RKD.

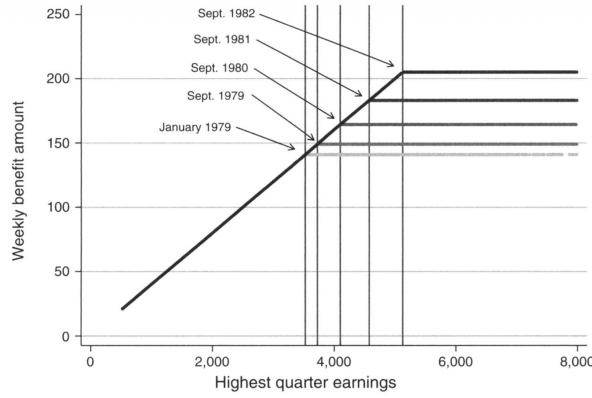


FIGURE 2. LOUISIANA: SCHEDULE OF UI WEEKLY BENEFIT AMOUNT, JAN. 1979–DEC. 1983

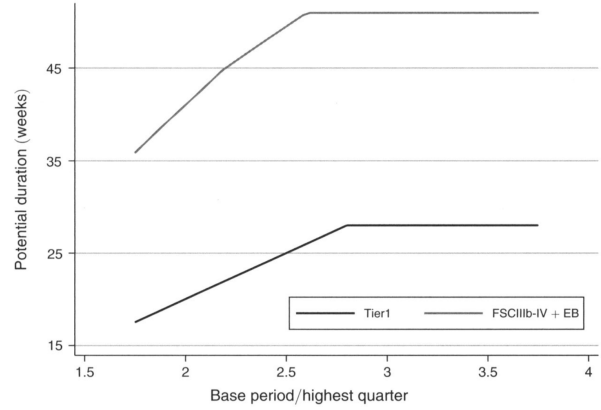


FIGURE 3. LOUISIANA: SCHEDULE OF UI POTENTIAL DURATION, JAN. 1979–DEC. 1983

Figure 3: Louisiana potential duration schedule over time

- Plots potential duration as function of bpw/hqw .
- Tier I schedule is kinked; during recessions, EB + federal extensions shift/extend the schedule and can move the kink.
- Interpretation: duration extensions create nonstationarity across time; empirically, I must restrict to spells where the duration schedule is stable over the relevant horizon.

Figure 4: RKD validity checks via density tests

- Panel A: density of hqw (normalized by kink) for benefit-level RKD. McCrary discontinuity estimate 0.067 (SE 0.059) and derivative discontinuity estimate 19.59 (SE 40.62) $\Rightarrow$ no evidence of manipulation or non-smooth density at kink.
- Panel B: density of bpw/hqw around kink for potential-duration RKD. McCrary discontinuity estimate -0.139 (SE 0.099) and derivative discontinuity estimate -216 (SE 220.6) $\Rightarrow$ again consistent with smoothness.
- Interpretation: supports the local random assignment / smoothness assumptions needed for RKD.

Figure 5: Covariate smoothness around the benefit-level kink (Louisiana)

- Plots binned means of covariates vs normalized hqw : age, education, capital income proxy, dependents.
- No visible kink/discontinuity in covariates at the benefit kink.
- Interpretation: reinforces that unobserved/observed composition is not shifting sharply at kink $\Rightarrow$ RKD is credible.

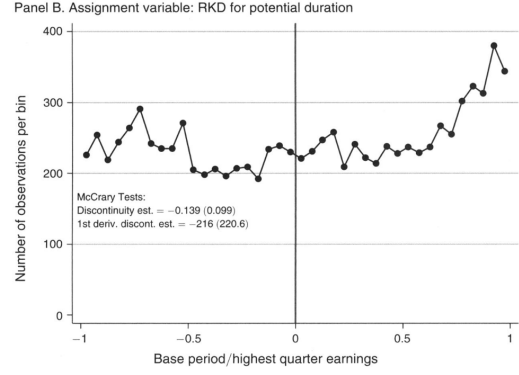
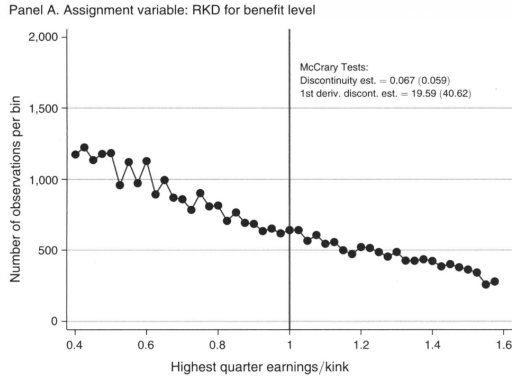


FIGURE 4. RKD GRAPHICAL EVIDENCE OF THE EFFECT OF UNEMPLOYMENT BENEFITS: DURATION OF UI CLAIMS

Figure 6: Graphical RKD evidence on outcomes (Louisiana)

- Panel A: duration of UI claims vs normalized hqw shows a visible kink at the benefit-level kink.
- Panel B: initial spell duration vs bpw/hqw (centered at kink) shows a kink at the potential-duration kink (noisier due to fewer observations).
- Interpretation: outcomes kink where policy rule kinks, consistent with a causal effect.

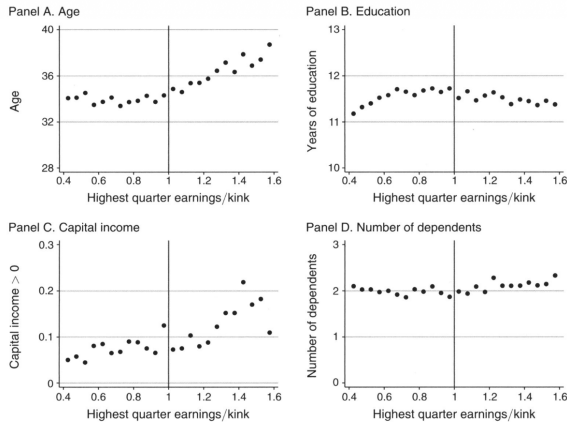


FIGURE 5. DISTRIBUTION OF HIGHEST QUARTER EARNINGS AND COVARIATES, LOUISIANA

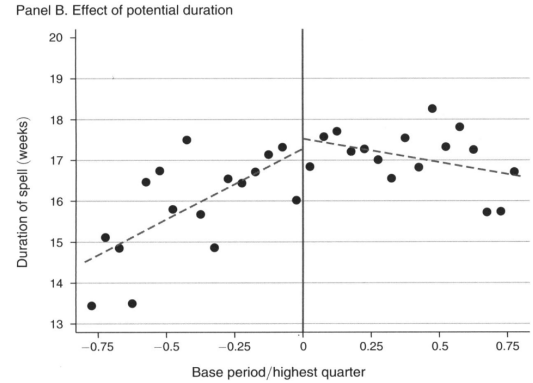
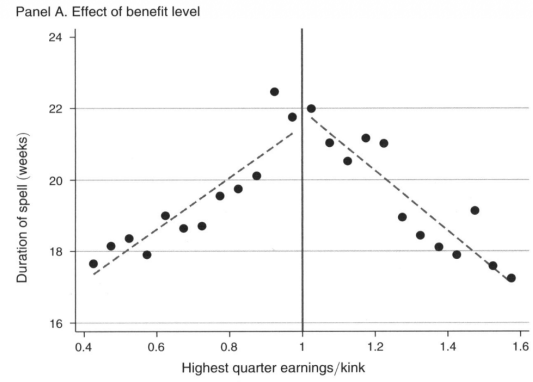


FIGURE 6. RKD GRAPHICAL EVIDENCE OF THE EFFECT OF UNEMPLOYMENT BENEFITS: DURATION OF UI CLAIMS, LOUISIANA 1979–1984

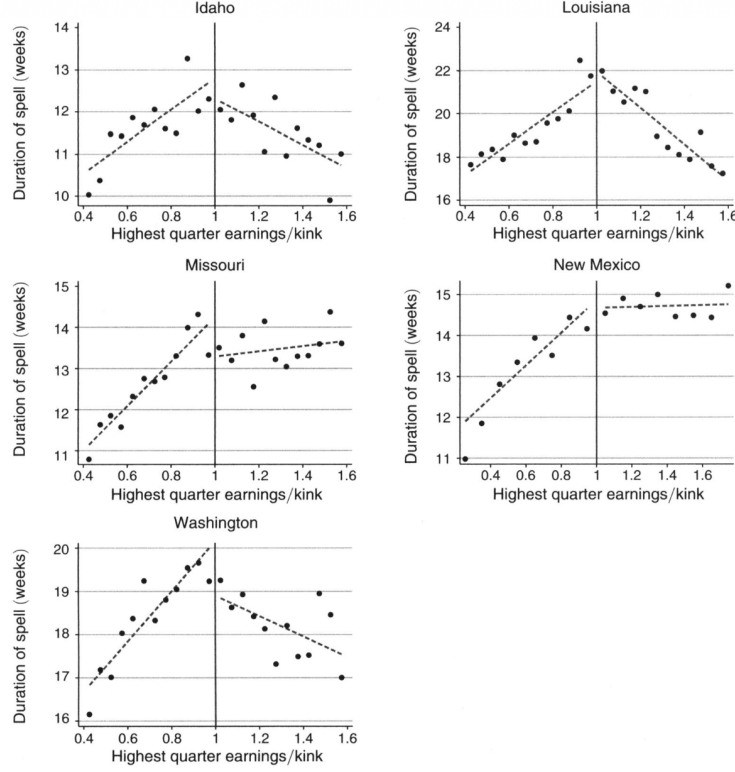


FIGURE 7. RKD EVIDENCE OF THE EFFECT OF BENEFIT LEVEL:
DURATION OF UI CLAIMS VERSUS HIGHEST QUARTER EARNINGS FOR ALL FIVE STATES

Figure 7: Benefit-level kink evidence across all five states

- For each state, duration of UI claims vs normalized hqw shows a kink at 1 (the kink point).
- Interpretation: replication across states reduces concern that a single-state functional form coincidence drives results.

4 Tables

Interpretation of Table 1.

- UI generosity differs a lot across states (weekly benefit wba ranges roughly from \$225–\$305 in \$2010; potential duration Tier I ranges from ~ 20 to ~ 27 weeks).
- There is substantial mass at maxima: e.g., Missouri has 64% at max weekly benefit; New Mexico has 92% at max potential duration.
- “Kink/avg. hqw ” varies a lot (e.g., 0.98 in Missouri vs 1.65 in Louisiana), which is helpful: kinks are not always at the same point of the earnings distribution across states, reducing functional-form coincidence concerns.
- Washington uniquely provides nonemployment duration (via wage records), enabling welfare calibration that needs both insured unemployment duration and total nonemployment.

Table 1 (part 1): Descriptive statistics for full CWBH sample — Idaho and Louisiana

	Idaho			Louisiana		
	Mean	SD	Obs.	Mean	SD	Obs.
<i>Duration outcomes (weeks)</i>						
Initial spell	13.8	12.3	33,125	17.5	14.1	44,702
Wks UI paid	11.7	10.7	33,125	17.3	13.8	44,702
Wks UI claim	15.8	12.2	33,125	18.7	13.7	44,702
<i>Earnings and benefits (\$2010)</i>						
<i>bpw</i>	25,163	22,227	33,125	26,894	19,029	44,702
<i>hqw</i>	9,835	16,463	33,125	9,538	6,228	44,702
<i>wba</i>	261.8	86.3	33,125	305.2	115.8	44,702
Pot. duration Tier I	20.0	5.5	33,125	24.9	4.3	44,702
Kink/avg. <i>hqw</i>	1.44	0.90	33,125	1.65	1.35	44,702
Percent with max. <i>b</i>	0.37	0.48	33,125	0.35	0.48	44,702
Percent with max. <i>D</i>	0.31	0.46	33,125	0.64	0.48	44,702
Average repl. rate	0.44	0.12	33,125	0.47	0.09	44,702
Exhaustion rate	0.11	0.29	33,125	0.13	0.31	44,702
<i>Covariates</i>						
Age	30.2	12.8	33,121	34.6	12.6	44,373
Male	0.67	0.47	33,121	0.70	0.46	44,058
Educ. (yrs)	12.0	2.2	17,627	11.4	2.7	41,308
Dependents	2.0	1.6	18,688	2.1	1.6	22,525

Table 1 (part 2): Descriptive statistics for full CWBH sample — Missouri and New Mexico

	Missouri			New Mexico		
	Mean	SD	Obs.	Mean	SD	Obs.
<i>Duration outcomes (weeks)</i>						
Initial spell	12.2	10.9	28,599	14.0	12.6	27,004
Wks UI paid	12.5	11.4	28,599	13.4	12.8	27,004
Wks UI claim	15.4	11.8	28,599	15.8	12.6	27,004
<i>Earnings and benefits (\$2010)</i>						
<i>bpw</i>	23,756	17,346	28,599	23,334	17,132	27,004
<i>hqw</i>	8,218	5,835	28,599	8,252	5,382	27,004
<i>wba</i>	224.9	51.4	28,599	230.0	69.5	27,004
Pot. duration Tier I	22.1	5.2	28,599	25.7	1.0	27,004
Kink/avg. <i>hqw</i>	0.98	0.74	28,599	1.30	0.80	27,004
Percent with max. <i>b</i>	0.64	0.48	28,599	0.43	0.50	27,004
Percent with max. <i>D</i>	0.52	0.50	28,599	0.92	0.27	27,004
Average repl. rate	0.45	0.16	28,599	0.43	0.11	27,004
Exhaustion rate	0.18	0.37	28,599	0.14	0.32	27,004
<i>Covariates</i>						
Age	34.8	12.7	28,585	33.7	11.4	26,924
Male	0.61	0.49	28,597	0.65	0.48	27,002
Educ. (yrs)	11.3	2.2	1,852	11.7	2.5	26,482
Dependents	2.0	1.6	21,701	2.2	1.7	25,534

Table 1 (part 3): Descriptive statistics for full CWBH sample — Washington

	Mean	SD	Obs.
<i>Duration outcomes (weeks)</i>			
Initial spell	17.6	15.4	41,992
Wks UI paid	16.2	14.8	41,992
Wks UI claim	18.9	15.4	41,992
Wks nonemployed	27.9	16.3	38,035
<i>Earnings and benefits (\$2010)</i>			
<i>b_{pw}</i>	31,232	20,380	41,992
<i>h_{qw}</i>	8,982	5,321	41,992
<i>w_{ba}</i>	286.7	94.7	41,992
Pot. duration Tier I	27.0	4.2	41,992
Kink/avg. <i>h_{qw}</i>	1.49	1.20	41,992
Percent with max. <i>b</i>	0.37	0.48	41,992
Percent with max. <i>D</i>	0.56	0.50	41,992
Average repl. rate	0.47	0.21	41,992
Exhaustion rate	0.12	0.31	41,992
<i>Covariates</i>			
Age	34.2	11.9	41,955
Male	0.627	0.484	41,972
Educ. (yrs)	12.4	2.4	41,702
Dependents	1.7	1.5	28,834

Interpretation of Table 2.

- α is the RKD estimate of the effect of a \$1 increase in weekly UI benefit on durations (in weeks). Across Louisiana subperiods, α is consistently positive for duration outcomes.
- Magnitudes: e.g. in 1981–1982, $\alpha \approx 0.04$ for paid UI duration $\Rightarrow$ \$1 more per week increases paid duration by about 0.04 weeks.
- Elasticities ϵ_b for duration outcomes are around 0.18–0.76 depending on subperiod: a 10% increase in WBA implies about a 2%–7% increase in unemployment duration locally (at the kink).
- Covariate columns (age, male, education, dependents) show no systematic kink (mostly tiny α), consistent with RKD validity.
- Reported p -values (goodness-of-fit comparisons with bin dummies) are generally not small for outcomes, suggesting the linear polynomial is not grossly misspecified locally (the paper treats this as supportive).

Interpretation of Table 3.

- β is the RKD estimate of the effect of **one additional week** of potential UI duration on outcomes (in weeks).

Table 2: RKD estimates of the effect of benefit level, Louisiana 1979–1983

	Duration of initial spell (1)	Duration UI claimed (2)	Duration UI paid (3)	Age (4)	Male (5)	Years of education (6)	Number of dependents (7)
<i>January–September 1979</i>							
α	0.006 (0.006)	0.007 (0.006)	0.006 (0.006)	−0.121 (0.069)	0.000 (0.002)	0.002 (0.014)	0.004 (0.010)
ϵ_b	0.183 (0.183)	0.228 (0.184)	0.186 (0.165)				
p -value	0.216	0.274	0.283	0.116	0.506	0.056	0.262
Observations	2,129	2,129	2,129	2,117	2,106	1,953	1,479
<i>September 1979–September 1980</i>							
α	0.018 (0.005)	0.019 (0.005)	0.018 (0.005)	0.052 (0.056)	−0.001 (0.002)	0.003 (0.012)	−0.001 (0.001)
ϵ_b	0.484 (0.143)	0.518 (0.142)	0.450 (0.131)				
p -value	0.589	0.389	0.499	0.659	0.041	0.735	0.742
Observations	3,765	3,765	3,765	3,752	3,723	3,483	2,042
<i>September 1980–September 1981</i>							
α	0.018 (0.006)	0.019 (0.006)	0.018 (0.006)	0.054 (0.069)	0.002 (0.000)	−0.025 (0.016)	−0.026 (0.010)
ϵ_b	0.455 (0.147)	0.467 (0.148)	0.422 (0.135)				
p -value	0.007	0.003	0.006	0.509	0.064	0.992	0.908
Observations	3,133	3,133	3,133	3,116	3,089	2,932	1,849
<i>September 1981–September 1982</i>							
α	0.042 (0.009)	0.038 (0.009)	0.040 (0.009)	0.051 (0.059)	−0.001 (0.002)	0.000 (0.014)	0.000 (0.014)
ϵ_b	0.708 (0.154)	0.665 (0.154)	0.644 (0.142)				
p -value	0.091	0.178	0.108	0.430	0.595	0.314	0.280
Observations	3,845	3,845	3,845	3,823	3,786	3,553	1,351
<i>September 1982–December 1983</i>							
α	0.047 (0.006)	0.046 (0.006)	0.042 (0.006)	−0.013 (0.005)	−0.001 (0.001)	0.001 (0.001)	−0.001 (0.001)
ϵ_b	0.757 (0.103)	0.763 (0.105)	0.667 (0.098)				
p -value	0.199	0.175	0.084	0.640	0.508	0.261	0.843
Observations	6,602	6,602	6,602	6,558	6,520	6,078	3,531

Table 3: Baseline RKD estimates of the effect of potential duration, Louisiana

	Duration of initial spell (1)	Duration UI claimed (2)	Duration UI paid (3)	Age (4)	Years of education (5)	Male (6)	Dependents (7)
<i>Period 1: January 1979–January 1980</i>							
β	0.210 (0.113)	0.184 (0.114)	0.211 (0.111)	−0.277 (1.609)	0.013 (0.030)	0.006 (0.006)	−0.027 (0.024)
ϵ_B	0.413 (0.223)	0.363 (0.225)	0.380 (0.200)				
Opt. poly	1	1	1	3	1	1	1
p -value	0.557	0.484	0.471	0.338	0.087	0.511	0.022
Observations	3,497	3,497	3,497	3,476	3,216	3,465	2,208
<i>Period 2: September 1981–April 1982</i>							
β	0.349 (0.141)	0.352 (0.138)	0.335 (0.136)	−0.251 (0.135)	0.005 (0.029)	0.002 (0.005)	−0.023 (0.030)
ϵ_B	0.793 (0.320)	0.804 (0.315)	0.710 (0.289)				
Opt. poly	1	1	1	1	1	1	1
p -value	0.133	0.149	0.107	0.486	0.493	0.842	0.388
Observations	2,165	2,165	2,165	2,148	1,959	2,138	888
<i>Period 3: June 1983–December 1983</i>							
β	0.387 (0.088)	0.363 (0.086)	0.334 (0.085)	−0.061 (0.079)	−0.014 (0.019)	0.006 (0.003)	−0.125 (0.061)
ϵ_B	0.854 (0.194)	0.851 (0.201)	0.708 (0.181)				
Opt. poly	1	1	1	1	1	1	2
p -value	0.675	0.751	0.742	0.624	0.898	0.493	0.754
Observations	2,936	2,936	2,936	2,917	2,720	2,904	1,601

- The estimated effects are about **0.2 to 0.4** additional weeks of unemployment duration per extra week of eligibility (across outcomes and subperiods).
- The implied elasticities ϵ_B are sizable (roughly 0.38 to 0.85 for duration outcomes). So duration extensions have strong effects on claim lengths.
- The paper emphasizes that the estimate contains a mechanical component $S(B)$ plus a behavioral component; since exhaustion rates are around 11%–18% in these data (Table 1), the mechanical effect alone cannot explain 0.2–0.4 weeks, implying meaningful behavioral responses.
- Covariate columns again show no robust kink, supporting identification.

Table 4: RKD estimates of behavioral responses to UI and liquidity/moral hazard estimates, Washington (July 1980–July 1981)

	Effect of benefit level (1)	Effect of potential duration (2)	Liquidity and moral hazard estimates (3)
ϵ_{DB}	0.730 (0.110) [0.814]	1.348 (0.685) [0.388]	
ϵ_D	0.291 (0.071) [0.392]	0.330 (0.425) [0.474]	
$(\frac{1}{B} \frac{\partial s_0}{\partial b} \Big _B - \frac{1}{b} \frac{\partial s_0}{\partial B}) \times 10^3$			−0.042 (0.010)
Moral hazard: Θ_1			0.0014 (0.0001)
Liquidity to moral hazard: ρ_1			0.876 (0.022)
Observations	6,061	2,049	9,471

interpretation of Table 4.

- Column (1): increasing **benefit level** increases insured unemployment duration quite a lot (elasticity $\epsilon_{DB} = 0.730$) and also increases total nonemployment duration (elasticity $\epsilon_D = 0.291$), consistent with longer jobless spells.
- Column (2): increasing **potential duration** also increases durations (UI duration elasticity even larger here: 1.348), though estimates are less precise (bigger SEs).
- Column (3): applies Proposition 1 to convert differential initial-exit responses into moral hazard Θ_1 and the **liquidity-to-moral-hazard ratio** ρ_1 .
- The estimated $\rho_1 = 0.876$ means liquidity effects are nearly as large as moral hazard effects:

$$\text{liquidity effect} \approx 0.876 \times (\text{moral hazard effect}).$$

So roughly, liquidity accounts for $\rho_1/(1 + \rho_1) \approx 0.467$ of the total response magnitude, while moral hazard accounts for ≈ 0.533 (if I interpret “total” as liquidity + moral hazard).

- Bottom line for a mental model: the data imply that UI lengthens unemployment not only because it reduces incentives (moral hazard) but also because it relaxes liquidity constraints (insurance value).

5 Big picture summary

- RKD evidence: duration outcomes kink where policy rules kink, while covariates and densities do not show kinks $\Rightarrow$ supports causal interpretation.
- Benefit level effects: elasticities of unemployment duration with respect to benefit level are typically in the 0.1–0.7 range (paper emphasizes an average around 0.3 for initial spell across many state-period estimates).
- Potential duration effects: each extra week of eligibility increases duration by about 0.2–0.4 weeks locally (Louisiana), implying substantial behavioral response beyond mechanical extension.
- Liquidity vs moral hazard: in Washington (1980–81), liquidity/moral hazard ratio is about 0.88, so liquidity effects are very important (almost one-for-one with moral hazard).
- Welfare calibration (qualitative, since the exact formulas are in the online appendix): given estimated elasticities and $\rho \approx 0.88$, the paper concludes that locally increasing benefit level and (slightly) increasing potential duration would be welfare-improving in that calibration setting.

Note. The paper repeatedly references **online appendix** tables/figures and the exact welfare formulas (e.g. formula numbers (31), (33) in the appendix). Those are not in the main PDF I’m using here, so I did not reproduce appendix-only tables or appendix-only model equations.

6 Conclusion

6.1 What is the paper ultimately concluding?

The paper’s bottom-line conclusions are:

- It is feasible to **separately identify moral hazard and liquidity effects** of unemployment insurance (UI) *using only administrative UI data*, by exploiting **time-profile variation** in benefits: changes in the *benefit level* b versus changes in *potential duration* B .
- Empirically (using a regression kink design on kinked UI schedules in five US states, late 1970s–1984), increases in UI generosity **significantly increase unemployment duration**, both for benefit level and for potential duration.

- The estimated labor-supply response is not “all moral hazard”: **liquidity effects are large**, with the **liquidity-to-moral-hazard ratio around 0.9** in the main calibration exercise (Washington, 1980–81).
- Under the paper’s (stylized) partial-equilibrium welfare calibration, these estimates imply that **a local increase in UI replacement rate (benefit level) and a small increase in potential duration can be welfare-improving** at the policy parameters where the sufficient statistics are measured.
- However, the paper stresses the key caveat that these welfare conclusions come from a **partial equilibrium** framework (infinitely elastic labor demand). In search-and-matching general equilibrium models, one would need **equilibrium employment responses** (e.g., job creation/search externalities) to fully assess optimal UI.

6.2 Summary of the paper (what it does, how, and what it finds)

This paper proposes a practical empirical method to evaluate the welfare effects of unemployment insurance by disentangling two distinct channels through which UI affects job search:

1. **Moral hazard (incentive/substitution) effect:** higher benefits reduce the net payoff to exiting unemployment quickly, lowering search effort.
2. **Liquidity (insurance) effect:** higher benefits relax cash constraints, allowing the unemployed to smooth consumption; this can also reduce search effort even holding incentives fixed.

The paper’s conceptual move is to connect the UI “moral hazard vs liquidity” decomposition (Chetty-style) to the **dynamic labor supply** insight (MaCurdy-style): by comparing the effect of changing returns *today* versus changing returns *in the future*, one can isolate the substitution component from the wealth component. In UI, the time profile of benefits provides the relevant “today vs future” variation:

- An increase in **benefit level b** shifts up benefits for all weeks of eligibility (front-loaded in the spell).
- An increase in **potential duration B** shifts benefits later (more back-loaded).

Because UI is a dynamic search problem with **state dependence** (future benefits matter only if the worker remains unemployed), the paper derives a correction using survival rates and durations (Proposition 1) so that the level-vs-duration comparison truly isolates moral hazard.

Empirically, the paper implements the strategy using a **Regression Kink Design (RKD)**: in many US states, UI benefit formulas are kinked deterministic functions of prior earnings. With high-quality administrative records (CWBH), the paper estimates:

- the elasticity of unemployment duration with respect to **benefit level**, and
- the effect (and elasticity) of unemployment duration with respect to **potential duration**.

Across states/periods, the paper finds sizeable behavioral responses. A concise summary of the headline magnitudes reported in the paper is:

- A **10% increase in benefit level** increases UI-claim duration by about **4% on average**.
- **One extra week** of potential duration increases UI-claim duration by about **0.3–0.4 weeks on average**.

Finally, using Washington (the only state where total nonemployment duration can be measured from wage records) and focusing on a period with stable schedules for both b and B , the paper estimates that **liquidity effects are almost as large as moral hazard effects** (liquidity/moral hazard ≈ 0.9) and concludes that small local increases in benefit level and duration would be welfare-improving under the paper’s calibration.

6.3 Intuition

The intuition I find is:

1) Why benefit level mixes moral hazard and liquidity. Raising the weekly benefit b makes unemployment less painful in two ways:

- it *reduces the incentive* to exit quickly (moral hazard),
- it *raises cash-on-hand/insurance* while unemployed (liquidity).

So the observed increase in duration from a higher b combines both channels.

2) Why potential duration is a “future benefits” shift. Extending B mainly shifts *extra benefits* to later in the spell (near exhaustion). That change still affects early search effort through expected future utility, but it does so in a way that is conceptually closer to a wealth/liquidity shift than an immediate incentive shift.

3) The key identification trick: equalize liquidity, compare timing. If I scale reforms so that they deliver the *same dollar liquidity* to someone who remains unemployed (heuristically $B db = b dB$), then:

- the liquidity component of a small level increase and a small duration extension is comparable,
- the *difference* in early-exit/search responses reveals the moral hazard component,

but because future benefits are received only if still unemployed (state dependence), I must correct the comparison using survival probabilities and average covered durations (Proposition 1).

4) Why the policy takeaway is not “cut UI to reduce duration.” The paper’s evidence says: yes, UI increases durations, but a large fraction of that response reflects **liquidity/insurance value** rather than pure distortion. Therefore, when doing welfare analysis, I cannot treat the duration increase as a pure deadweight loss; the liquidity channel is part of the welfare gain from insurance.